

LESSON OVERVIEW

Recommended length: 2-3 class periods

Students reverse-engineer a small purpose-made assembly and produce a technical documentation package that integrates isometric sketching, multiview drawing, line conventions, dimensioning, measurement, tolerances, joining methods, and evidence-based improvement.

SUGGESTED ASSEMBLY: AEROSPACE SENSOR MOUNT MOCK-UP

Reusable and purpose-made

- Base plate:** 3.000 x 1.500 x 0.250 in; two mounting holes; one attachment interface
- Upright bracket:** 1.500 x 1.250 x 0.250 in vertical plate; sensor hole or slot; attaches to base
- Spacer or locator block:** 0.500 x 0.500 x 0.250 in; creates offset or alignment feature
- Joining hardware:** Use screws, bolts/nuts, or tab/slot plus fastener depending on the interface you want students to analyze

CLASSROOM FLOW

Keep most work student-driven

- Day 1:** Introduce challenge, identify parts/interfaces, complete assembly snapshot, isometric sketch, and view planning.
- Day 2:** Measure parts, create multiview documentation, add dimensions, identify critical dimensions, and assign tolerances.
- Day 3:** Complete joining/interface analysis, propose improvement, peer review, revise, and submit.

MATERIALS

One assembly per team is ideal

- student packet
  - rulers and calipers
  - pencils and straightedges
- purpose-made assemblies
  - plain/isometric graph paper
  - optional: spare fasteners or alternate joints

## ASSESSMENT LOOK-FORS

Use this to grade efficiently

|                            |                                                                                          |
|----------------------------|------------------------------------------------------------------------------------------|
| <b>Sketching/views:</b>    | Views are understandable, aligned, and use object, hidden, and center lines correctly.   |
| <b>Dimensions:</b>         | Key sizes and feature locations are present. Students avoid repeated dimensions.         |
| <b>Measurement:</b>        | Reported values are realistic for the tool used and include units.                       |
| <b>Tolerances:</b>         | Chosen tolerances connect to fit, alignment, function, or manufacturing.                 |
| <b>Interface analysis:</b> | Students identify both parts in a joint and explain what the connection must accomplish. |
| <b>Improvement:</b>        | The proposed change improves a documented issue or tradeoff.                             |

## COMMON STUDENT ISSUES

Use for quick feedback

- Missing location dimensions for holes or slots
- Only drawing one view when another view is needed
- Repeating the same dimension in multiple places
- Writing tolerance values without explaining why they matter
- Calling every connection strong without addressing serviceability or assembly
- Suggesting an improvement that is not connected to evidence

## SAMPLE FEEDBACK PHRASES

- Your views communicate the shape well, but the hole/slot still needs a location dimension.
- Your measurement record is strong; now connect one tolerance to a specific fit or alignment requirement.
- Your joining analysis identifies the hardware, but it also needs to explain what the interface must do.
- Your improvement is realistic. Add evidence from your measurements or interface analysis to support it.